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A new diterpene from the fungal mycelia of *Hericium erinaceus*

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ABSTRACT

One new diterpene (**1**) and new compound (**2**), along with four known compounds (**3**–**6**) were isolated from the fungal mycelia of *Hericium erinaceus* by the tracking method of antibacterial activity. The structures of compounds (**1**–**6**) were elucidated on the basis of spectral data. Compound (**3**) showed strong antibacterial activity against *Helicobacter pylori* (ATCC43504) and compounds (**1**), (**2**) and (**4**) showed moderate antibacterial activity. Compound (**1**) exhibited good cytotoxicity against three human cancer cell lines (K562, LANCAP, HEP2).

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1. Introduction

Mushrooms are widely used as food materials with their unique flavor and have also been used in oriental medicine because of their beneficial effects such as antitumor and antibacterial properties (Kodama et al., 2003; Liu et al., 2010). *Hericium erinaceus*, a well-known edible and medical mushroom, has been used as a traditional Chinese medicinal fungus for many years. Previous studies reported that both the fruiting bodies and the fungal mycelia of *H. erinaceus* contain various bioactive constituents such as polysaccharides, lectins, proteins, hericenones, erinacines, erinapyrones, sterols, fatty acid and esters (Han et al., 2013; Kawagishi et al., 1992, 1994a,b; Kenmoku et al., 2001; Lee et al., 2000). Accumulating data suggest that *H. erinaceus* has beneficial effects, including anticancer activity (Mizuno et al., 1992), antibacterial activity, anti-inflammatory activity (Kim et al., 2012), antioxidant properties (Zhang et al., 2006) and stimulating the synthesis of nerve growth factor (Kawagishi et al., 1991, 1994a,b). In order to further discovering more bioactive compounds from *H. erinaceus*, we report herein the isolation and structure elucidation of a new diterpene (**1**) and one new compound (**2**), together with four known compounds (**3**–**6**) from the fungal mycelia of *H. erinaceus* by the tracking method of antibacterial activity. The antibacterial activity against *Helicobacter pylori* (ATCC43504) and antineoplastic activity against three human cancer cell lines (K562, LANCAP, HEP2) were also evaluated in the course of gaining an insight into the beneficial effects of the mushroom *Hericium erinaceum*.

2. Results and discussion

The dried fungal mycelia of *H. erinaceus*, which was dissolved in H_2O, was absorbed by macroporous resin HPD-100 and eluted by EtOH with different concentration. Since the fraction eluted by 70–100% aq. EtOH exhibited significant antibacterial activity, this fraction was extracted with petroleum ether, Et_2O, $CHCl_3$ and *n*-BuOH. The Et_2O soluble fraction was purified by chromatographic separation over silica gel, yielding the new diterpene (**1**) and two known compounds identified as 4-chloro-3,5-dimethoxybenzyl alcohol (**4**) (Okamoto et al., 1993) and 7α, 8β, 11-trihydroxydrimane (**6**) (Liu et al., 2008) (Fig. 1). The purification of the $CHCl_3$ soluble fraction gave the new compound 3-acetyl-4-methoylbenzoic acid (**2**), together with two known compounds: 1-(5-chloro-2-hydroxyphenyl)-3-methyl-1-butanone (**3**) and 4-chloro-3, 5-dimethoxybenzoic acid (**5**) (Qian et al., 1989). Analysis on petroleum ether soluble fraction and *n*-BuOH soluble fraction are under progress in our laboratory.

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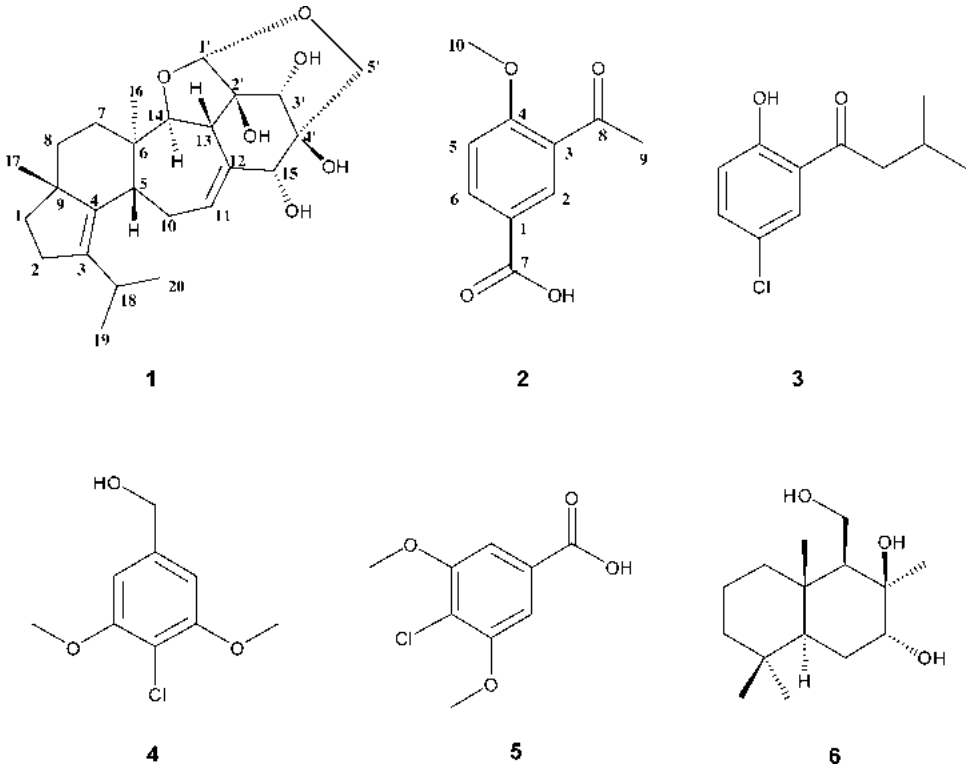

Fig. 1. The structures of compounds **1**–**6**.

2.1. Structure of elucidation of compounds 1–2

Compound **1** was isolated as white powder. The molecular formula $C_{25}H_{36}O_6$ was established by the HREIMS ion at *m/z* 432.2516 ($[M]^+$, calcd 432.2512). The ^{1}H NMR spectra of compound **1** showed the presence of thirty-two protons and decreased four protons comparing with the molecular formula, so compound **1** could contain four hydroxyl groups. The ^{1}H NMR spectrum data (Table 1) of **1** gave resonances for an olefinic proton at δ 5.71 (1H, t, J = 3.2 Hz), six oxymethines at δ 4.98 (1H, s), 4.41 (1H, s), 4.24 (1H, d, J = 11.9 Hz), 3.61 (1H, d, J = 1.2 Hz), 3.72 (1H, d, J = 1.4 Hz), 3.67 (1H, dd, J = 7.5 Hz, 1.4 Hz), four methyl groups at δ 1.08 (3H, s), 1.02 (3H, s), 0.98 (3H, d, J = 6.0 Hz), 0.96 (3H, d, J = 6.0 Hz).

The ^{13}C NMR and DEPT spectrum data (Table 1) of compound **1** displayed four methyl groups at δ 18.0, 22.1, 22.6, 25.4, six methylenes at δ 28.7, 29.4, 31.4, 37.7, 39.6, 65.6, eight methines, and seven quaternary carbons. Compound **1** involved the following functional groups: two double bonds at δ 129.4, 139.6, 140.6, 141.3; a hemiketone carbon at δ 109.2; a oxymethylene carbon at δ 65.6; three oxymethine carbons at δ 92.0, 85.2, 71.4; two quaternary oxygenated carbons at δ 87.6, 78.9. On the basis of these data, compound **1** was initially presumed to be a diterpene having "cyathan" skeleton like erinacines E or F (a diastereomer of erinacine E) (Kawagishi et al., 1996).

The basic "cyathan" skeleton of compound **1** was confirmed by the following key HMBC correlations H-5/C-3, C-4, C-10, C-11; H-10/C-11; H-14/C-6, C-7, C-12; H-15/C-3′, C-4′, C-11, C-13; H-16/C-5, C-6, C-7, C-14; H-17/C-1, C-4, C-8, C-9; H-20/C-3, C-19; H-1′/C-14, C-13; H-3′/C-2′, C-15; H-5′/C-1′, C-3′ (Fig. 2). The relative configurations of all the chiral centers of **1** and the conformation of each ring were elucidated by analysis of the ^{1}H NMR spectra and the ROESY spectrum (Fig. 2).

In the ^{1}H NMR spectra, the coupling constant between H-13 and H-14 (J = 11.9 Hz) of **1** indicated that their relative configuration was trans. In the ROESY spectrum, the relative stereochemistry of the aglycone part in **1** was the same as that in erinacine E (Kawagishi et al., 1996). The observed ROESY cross peaks of H-5/H-17, H-5/H-13, H-13/H-15, H-3′/H-15, suggested that they were cofacial and β-oriented. The strong ROESY correlations of H-14/H-16 indicated that they were in α-orientation. The above deduction, combined with the key HMBC correlations led us to draw the conclusion that compound **1** was a diastereomer of erinacine E in the sugar part (Fig. 2).

The absolute configuration of the sugar part in compound **1**, from 1′, 2′ and 4′ uncertain chiral center, was determined by

Table 1
^{1}H and ^{13}C NMR spectroscopic data of compounds **1** in CD_3OD and **2** in $(CD_3)_2CO$.

Position	δ_H (J Hz), **1**	δ_C, **1**	δ_H (J Hz), **2**	δ_C, **2**
1	1.61 (m), 1.56 (m)	39.6		123.6
2	2.32 (m), 2.28 (m)	29.4	8.31 (d, 2.9)	132.6
3		140.6		129.1
4		139.6		163.2
5	2.71 (d, 9.1)	44.6	7.27 (d, 10.9)	113.1
6		42.3	8.15 (dd, 10.9, 2.9)	135.8
7	1.81 (m), 1.61 (m)	28.7		166.8
8	1.61 (m), 1.58 (m)	37.7		198.4
9		51.3	2.56 (s)	56.6
10	2.62 (d, 5.3), 2.41 (d, 7.4)	31.4	4.05 (s)	31.8
11	5.71 (t, 3.2)	129.4		
12		141.3		
13	3.48 (d, 11.9)	56.2		
14	4.24 (d, 11.9)	92.0		
15	4.41 (s)	85.2		
16	1.02 (s)	18.0		
17	1.08 (s)	25.4		
18	2.87 (m)	28.6		
19	0.98 (d, 6.0)	22.1		
20	0.96 (d, 6.0)	22.6		
1′	4.98 (s)	109.2		
2′		78.9		
3′	3.61 (d, 1.2)	71.4		
4′		87.6		
5′	3.72 (d, 1.4), 3.67 (dd, 7.5, 1.4)	65.6		

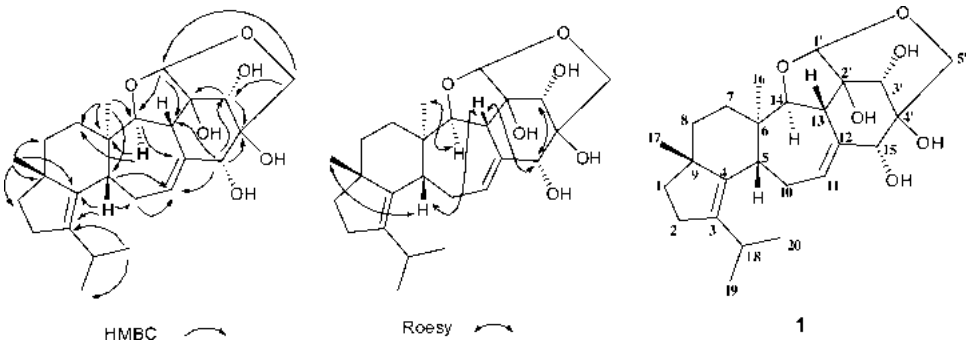

Fig. 2. The key HMBC and ROESY correlations of compound **1**.

comparison of calculated and experimental Vibrational Circular Dichroism (VCD) spectra (Fig. 3). There are eight possible stereoisomers of compound **1**, here abbreviated as RRR, RRS, RSS, RSR, SSS, SSR, SRS, and SRR (where, e.g., RRS = 1′R, 2′R and 4′S). All isomers were optimized quantum mechanically at the B3LYP/CV-PVTZ theory level. Analysis of the potential energy surfaces revealed local minima with relative enthalpies within 1.4 kcal/mol. For each isomer, all conformers were therefore taken into account in calculations of the spectral properties. When compared with the experimental VCD spectrum, it can be found that the isomer SRS spectra conformed to the highest degree and the energy value of the configuration is the lowest (Fig. 3). Thus the absolute configuration of 1′, 2′ and 4′ chiral center in compound **1** was determined as 1′S, 2′R and 4′S (Fig. 1).

Compound **2** was also obtained as white powder and the molecular formula $C_{10}H_{10}O_4$ was established by the HREIMS ion at m/z 194.0578 ([M]$^+$, calcd 194.0579). In the ^{13}C NMR spectrum of compound **2**, six carbon signals were displayed at δ 113.1, 123.6, 129.1, 132.6, 135.8, 163.2, and the corresponding to proton signals were observed at δ 7.27 (d, J = 10.9 Hz), 8.15 (dd, J = 10.9 Hz, 2.9 Hz), 8.31 (d, J = 2.9 Hz), suggesting an aromatic compound with three substituents. Compound **2** contained the following functional groups: a methyl carbon at δ 31.8; an oxymethyl carbon at δ 56.6; a carbonyl carbon at δ 166.8; a ketone carbon at δ 198.4. In the HMBC spectrum, the oxymethyl protons at δH 4.05 (H-10) showed a cross-peek with a quaternary carbon δC 163.2 (C-4), suggesting that the oxymethyl group was placed at C-4 of the benzene ring. The acetyl unit was attached to the benzene ring at C-3 according to HMBC correlations of the protons δH 2.56 (H-9) with δC 129.1 (C-3). The following key HMBC correlations H-2/C-4, C-6, C-7, C-8; H-5/C-1, C-3; H-6/C-2, C-4, C-7, suggesting that the carboxyl group was located at C-1 of the benzene ring. The above deduction, combined with the key HMBC correlations led to the establishment of the structure of compound **2** as 3-acetyl-4-methoylbenzoic acid.

2.2. Antibacterial activity and cytotoxicity

The antibacterial activity of the ethanol eluent with different concentration was tested by a modified version of the microplate agar diffusion method against *H. pylori* (ATCC43504) (Kobayashi et al., 2004). 70% ethanol eluent inhibited the growth of the *H. pylori* (ATCC43504) test strain with MIC values 250–500 μg/mL; 90% and 100% ethanol eluent were also relatively inhibitory to

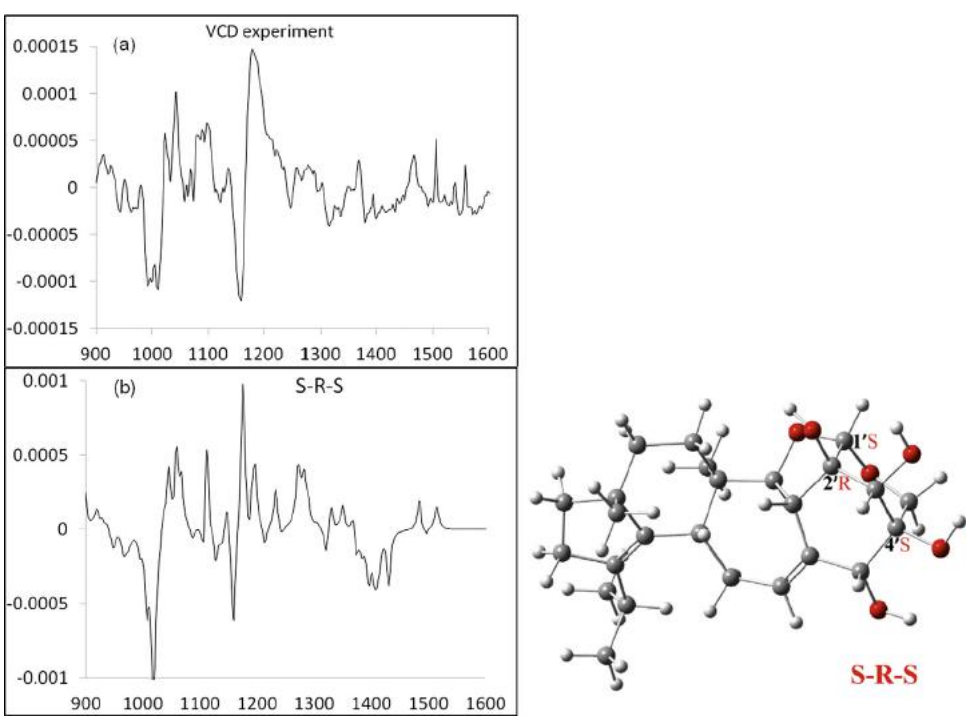

Fig. 3. (a) Experimental VCD spectrum of compound **1**, and (b) computed VCD spectra for the isomers (1′S, 2′R, 4′S) of compound **1**.

Table 2
Minimal inhibitory concentration (MIC) from *H. erinaceus* against *Helicobacter pylori* (ATCC43504).

Ethanol eluent with different concentration	MIC (μg/mL)	Fraction	MIC (μg/mL)	Compound	MIC (μg/mL)
0	0	[a]Ethanol eluent	125–250	**1**	50–100
10%	>800	Petroleum ether	125–250	**2**	50–100
30%	>1760	Et_2O	125–250	**3**	32.5–65
50%	>1200	$CHCl_3$	50–100	**4**	62.5–125
70%	250–500	*n*-BuOH	25–50	**5**	125–250
90%	110–220			**6**	>2000
100%	240–280				

[a] Ethanol eluent: 70–100%.

H. pylori (ATCC43504) (MIC values 110–220 μg/mL and 240–480 μg/mL) (Table 2). Therefore, dried fungal mycelia of *H. erinaceus* were eluted with 70–100% ethanol after absorbed by macroporous resin HPD-100.

The effects of petroleum ether soluble fraction, Et_2O soluble fraction, $CHCl_3$ soluble fraction, *n*-BuOH soluble fraction, and compounds **1**–**6** against the growth of *H. pylori* (ATCC43504) are all shown in Table 2. $CHCl_3$ soluble fraction and *n*-BuOH soluble fraction showed moderate antibacterial activity against *H. pylori* (ATCC43504) (MIC values 50–100 μg/mL and 25–50 μg/mL). Compound **3** which isolated from $CHCl_3$ soluble fraction performed strong antibacterial activity against *H. pylori* (ATCC43504) with MIC values 32.5–65 μg/mL, suggesting that compound **3** could be the main component of antibacterial activity in $CHCl_3$ soluble fraction. Compound **1** obtained from Et_2O soluble fraction showed moderate antibacterial activity against the growth of *H. pylori* (ATCC43504) (MIC values 50–100 μg/mL), suggesting that compound **1** could be the main component of antibacterial activity in Et_2O soluble fraction. Both compound **2** and **4** showed relatively moderate antibacterial activity, while compound **6** expressed no antibacterial activity.

Compounds **1**–**6** were evaluated for their cytotoxicity against three human tumor cell lines including the human erythroleukemia cell line K562, the human prostate cancer cell line LANCAP and the human laryngeal epithelial cell line HEP2 by the MTT method reported (Alley et al., 1988). Compound **1** showed good cytotoxicity against two of the above-mentioned tumor cell lines (K562 and HEP2) with $IC_{50} < 200$ μM, and compound **2** showed no cytotoxicity against all the assayed cell lines (Table 3). The $IC_{50} > 800$ μM were calculated on all the assayed cell lines for Compounds **3**–**6**, suggesting that compounds **3**–**6** exhibited weak cytotoxicity.

3. Experimental

3.1. General experimental procedures

Infrared (IR) spectra were recorded on a Bruker Vector 22 spectrometer with a KBr pellet. America Bio Tools company 1D NMR (^{1}H NMR and ^{13}C NMR) and 2D NMR (ROESY, HMBC) spectra were recorded on a Bruker Avance 500 NMR spectrometer operating at 500 MHz (^{1}H) and 125 MHz (^{13}C). Chemical shifts were given in ppm and tetramethylsilane (TMS) was used as an internal standard. HREIMS and ESIMS spectra were recorded on Waters Micromass GTC spectrometer (Waters Technologies, USA). Sephadex LH-20 (Pharmacia, Uppsala, Sweden), Silica gel (Sinopharm Chemical Reagent Co. Ltd., Shanghai, China, 100–200 and 200–300 mesh) and YMC-Gel RP-C18 (50 μm, YMC, USA) were used for column chromatography. TLC was carried out on precoated silica gel GF254 plates (Qingdao Haiyang Chemical Co. Ltd., Qingdao, China), and spots were detected by heating after spraying with vanillin–sulfuric acid. Macroporous resin HPD-100 (Baoen Chemical Company, Hebei Province, China) was used for absorbing active compounds. Preparative HPLC was carried out using LC-3000 with shim-pack RP-C18 column (25 mm × 300 mm, i.d., 5 μm) and binary channel UV detector (Beijing Chuangxin Tonghen Science Co. Ltd., Beijing, China). Analysis HPLC was performed using a Waters 2695 HPLC system (Waters, USA) equipped with a quaternary solvent delivery system, an autosampler, an ultraviolet detector and a Parallax detector. An ODS C18 column was used for analytical (4.6 mm, 250 mm; Waters, USA) purposes. DMEM and fetal calf serum (FBS) (Gibco BRL, France), and 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyltetrazolium bromide (MTT) were used for the test of antitumor activity. All solvents used were analytical grade (Sinopharm Chemical Reagent Co. Ltd., Shanghai, China).

Table 3
Cytotoxicity of compounds **1**–**6**, preformed as IC_{50} in μM.

Samples	K562	LANCAP	HEP2
1	124.5 ± 1.9	362.3 ± 6.7	198.6 ± 3.6
2	–	–	–
3	930.2 ± 2.5	>2000	1368.4 ± 3.1
4	1491.1 ± 3.8	>2000	1474.1 ± 4.3
5	1638.1 ± 4.1	>2000	1403.9 ± 5.2
6	1394.9 ± 3.9	848.0 ± 7.1	>2000
Fluorouracil[a]	76.9 ± 1.4	61.5 ± 2.6	38.5 ± 2.9

–: not tested.
[a] Positive control.

3.1.1. VCD measurement

VCD measurements were performed using a Chiral IR-2X TM Vibrational Circular Dichroism spectrophotometer equipped with dual photoelastic modulation. Compound **1** (6.8 mg) in CH_3OH (150 μL) was placed in a BaF_2 cell with a path length of 100 μm and data was acquired at a resolution of 4 cm^{-1}.

3.1.2. Computational details

Eight relevant configurations, from 1′, 2′ and 4′ uncertain chiral center, were taken into account. Only one conformation was performed for each configuration, because of rigidity of rings in the compound. The geometries were fully optimized in the vacuum without any constraints at B3LYP/CV-PVTZ theory level, all calculation was performed with Gaussian 09 software package. The theoretical VCD spectra were obtained from the GaussView software fitting with the experimental step-size after it calculated at the same theory level.

3.2. Fungus materials and cultivation

The fungal mycelium of *H. erinaceum* was prepared as follows: *H. erinaceus* strain 0605 (provided by the Institute of Edible Fungi, Shanghai Academy of Agricultural Sciences, Shanghai, China) was cultured on potato dextrose agar (Difco Laboratories, Detroit, MI, USA) slants or plates at 28 °C for 12 days. Cultures

were suspended in 100 mL sterile distilled water and homogenized in a Waring blender operated for 15 s at full power. Homogenate (15 mL) was used to inoculate 200 mL of potato dextrose broth (Difco Laboratories) in 250-mL Erlenmeyer flasks, and cultures were incubated in the dark with shaking (180 rpm) at 28 °C for 14 days. The dry fungal mycelium was obtained by filtration through sterile gauze and was dried by vacuum dryer at −60 °C (Shang et al., 2013).

3.3. Isolation

A total of 2.5 kg of dried and powdered fungal mycelium of *H. erinaceum* was suspended in boiling water (25 L) and stirred for 2 h. After cooled to the room temperature, 5 kg of macroporous resin HPD-100 was added and stirred for 24 h. The macroporous resin HPD-100 was subjected to a glass column (ø 10.0 cm × 80.0 cm), eluted with water (15 L), 50% ethanol (20 L), 70% ethanol (20 L), and 100% ethanol (20 L). The elution of 70–100% ethanol was collected and concentrated to crude extract (532 g). A part of the crude extract (250 g) was suspended in H_2O and successively partitioned with petroleum ether, Et_2O, $CHCl_3$ and *n*-BuOH (3 × 300 mL, each), respectively, obtaining a petroleum ether soluble fraction (2.31 g), a Et_2O soluble fraction (12.24 g), a $CHCl_3$ soluble fraction (26.74 g) and a *n*-BuOH soluble fraction (42.79 g). The Et_2O soluble fraction (12.24 g) was subjected to a silica gel column (200–300 mesh), using a gradient of petroleum ether–EtOAc (49:1–1:1) to give 18 fractions. The fraction 11 (80.5 mg) was purified by preparative HPLC to afford compound **4** (10 mg, t_R = 14.0 min), eluted with MeOH–H_2O (40:60) as the mobile phase. The fraction 15 (112.6 mg) was separated on an RP-C_{18} silica gel column to give compound **1** (20 mg), eluted with MeOH–H_2O (50:50, 80:20) as the mobile phase. The fraction 14 (258.6 mg) was recrystallized in EtOAc to obtain compound **6** (116 mg). The $CHCl_3$ soluble fraction (26.74 g) was chromatographed to a silica gel column (200–300 mesh), eluted a gradient of $CHCl_3$–actone (95:5–50:50) to give 14 fractions. The fraction 3 (98.6 mg) was recrystallized in acetone to obtain compound **3** (31 mg). The fraction 5 (185.3 mg) and the fraction 9 (149.7 mg) were separated respectively on a Sephadex LH-20 column to give compound **5** (89 mg) and compound **2** (12 mg), eluted with MeOH–H_2O (50:50, 90:10) as the mobile phase.

3.3.1. Compound *1*

White powder; IR (KBr), ν_{max} cm^{-1}: 3443, 2928, 2864, 1637, 1453, 1381, 1072, 1027; ^1H and ^{13}C NMR, see Table 1; HREIMS (positive-ion mode) *m/z*: 432.2516 $[M]^+$ (calcd for $C_{25}H_{36}O_6$: 432.2512); ESIMS (negative-ion mode) *m/z*: 431.2 $[M-H]^-$, 467.2 $[M+Cl]^-$; $[\alpha]_D^{20}$ −87.2° (c 1.10, C_2H_5OH).

3.3.2. Compound *2*

White powder; IR (KBr), ν_{max} cm^{-1}: 3444, 3031, 2590, 1706, 1637, 1602, 1247, 1218, 1017, 592; ^1H and ^{13}C NMR, see Table 1; HREIMS (positive-ion mode) *m/z*: 194.0578 $[M]^+$ (calcd for $C_{10}H_{10}O_4$: 194.0579); ESIMS (positive-ion mode) *m/z*: 194.9 $[M+H]^+$.

3.4. Antibacterial activity

American Type of Culture Collection (ATCC) standard against namely: *H. pylori*. *H. pylori* strains used to screen antibacterial activity were standard reference strains ATCC 43504. Strains were maintained in skimmed milk at −76 °C for long-term preservation (Han et al., 1995).

The microplate agar diffusion assay was used with some modifications (Rasoanaivo and Ratsimamanga-Urveg, 1993; NCCLS, 1997; Kobayashi et al., 2004) to determine the minimum inhibitory concentration (MIC) values of ethanol eluent, the petroleum ether soluble fraction, Et_2O soluble fraction, $CHCl_3$ soluble fraction, *n*-BuOH soluble fraction, and compounds **1**–**6**. Two-fold serial dilutions (50 μL) of *H. erinaceus* tested samples (in ethanol) were transferred to each well of a 96-well microplate. Ethanol had been added to wells and then air-dried served as negative controls. After sterilizing the well plates by 15 min of exposure to ultraviolet irradiation, aliquots (100 μL) of molten brucella broth modified agar containing 7–10% sheep blood (*H. pylori*) were added to the wells. Then wells were inoculated with the test bacteria using sterile swabs, and the cultures were incubated at 37 °C for 3 days under a microaerophilic atmosphere (*H. pylori*). MIC values were defined as the lowest concentration that completely inhibited visible bacterial growth.

3.5. Cytotoxicity

Human erythroleukemia cell line K562, human prostate cancer cell line LANCAP and human laryngeal epithelial cell line HEP2 were purchased from American Type of Culture Collection (ATCC). Cells were grown in DMEM, supplemented with 10% PBS, 100 mg/mL of streptomycin, 100 units/mL of penicillin. Cells were incubated under 5% CO_2 humidified atmosphere at 37 °C.

The effect of compounds **1**–**6** from *H. erinaceum* on cell viability was assessed by the MTT assay. The K562, LANCAP and HEP2 cells were plated in 96-well plates at concentration of 10^5 cells/well and incubated for 24 h, then treated with various concentrations of compounds **1**–**6**, which diluted with culture medium to give the appropriate final concentrations of 12.5–200 μg/mL. 5-Fluoro-2,4-(1h, 3h)-pyrimidinedione (5-FU) used as positive control which were dissolved in dimethyl sulfoxide (DMSO). Controls were done with the same final DMSO concentration in the medium as samples. After incubation of the cells with compounds **1**–**6** for 48 h at 37 °C, 20 mL of MTT (5 mg/mL) solution were added to each well and incubated for 4 h at 37 °C. Then, the broth was removed and DMSO (150 μL) was added to each well. The degree of formazan formation, an indicator of living and metabolically active cells, was determined using ELISA reader at 570 nm. Time point was measured in five different experiments.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.phytol.2014.12.011.

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